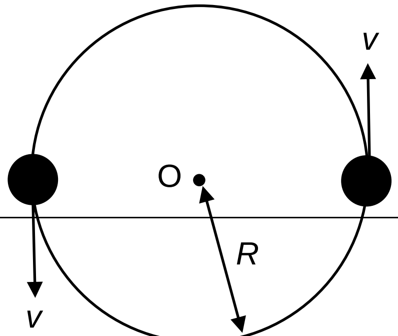
10 Two stars of equal mass M move with constant speed v in a circular orbit of radius R about their common centre of mass O as shown.



What is the magnitude of the centripetal force on each star?

A	$\dfrac{GM^2}{4R^2}$	**B**	$\dfrac{GM^2}{R^2}$	**C**	$\dfrac{2Mv^2}{R}$	**D**	$\dfrac{Mv^2}{2R}$

Answer: A

Using Newton's law of gravitation, $F = \dfrac{Gm_1m_2}{r^2}$ where m_1 and m_2 are the respective masses and r is the distance between the 2 masses;

$$F = \frac{GMM}{(2R)^2} = \frac{GM^2}{4R^2}$$

11 The planet Jupiter has a radius of 7.15×10^7 m and is approximately 318 times more massive